Thu Htet

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EDUCATION

University of California, Davis | GPA 3.42 **Davis, CA**
BACHELOR OF SCIENCE – ELECTRICAL & COMPUTER ENGINEERING **Expected June 2026**
Circuits, VLSI Design, Digital Systems, Digital Electronic Circuits (CMOS), Semiconductor Physics, Embedded Systems, Testing and Verification of Digital Systems, Digital Signal Processing, Computer Architecture, Computer Networks, Algorithm Analysis

Teaching Assistant for Java, C++, Data Structures, Algorithms, and Object-oriented Programming

HARDWARE/FIRMWARE PROJECTS

Smart Home Automation System - PCB Design & Embedded Systems **March 2026 - June 2026**
*** UC Davis ECE Outstanding Senior Design Project Award ***

- Designed a modular smart home control interface featuring buttons, rotary encoders, and magnetic expansion modules for configurable device control
- Developed a compact PCB integrating a microcontroller, user input components, and external interfaces to support flexible expansion with additional modules
- Implemented I2C-based communication between microcontroller and peripheral modules for real-time control and status monitoring.

Satellite Battery Management System **December 2024 - November 2025**

- Designed a 4S lithium-ion battery management system featuring battery protection, cell monitoring, and power management functions for CubeSat applications
- Created schematics and PCB layouts in KiCad, integrating protection ICs, sensing circuitry, and power distribution networks while following PCB design best practices
- Built and validated hardware prototypes through soldering, debugging, and environmental testing, including thermal, short-circuit, and vibration evaluations

Smart Fridge Device — Embedded Firmware & Circuit Testing **February 2026**

- Developed embedded C firmware on a PSoC microcontroller to monitor refrigerator door status and temperature, generating real-time visual and audio alerts
- Integrated thermal and magnetic sensors with ADC and GPIO peripherals to enable real-time environmental monitoring and event detection
- Configured and debugged I2C, GPIO, timers, and ADC peripherals in PSoC Creator, validating hardware–software interaction on prototype hardware

Matrix Multiplier — Verilog RTL and FPGA Digital Design **October 2024**

- Designed and implemented a matrix multiplication accelerator in Verilog, including multiply-accumulate (MAC) datapaths, RAM blocks, and controller logic using ModelSim and Intel Quartus Prime
- Developed and verified RTL modules and FSM-based control logic through simulation, waveform analysis, and testbench validation
- Improved throughput by implementing dual-MAC architectures, pipelining, and optimized memory access patterns, reducing clock cycles per output operation

RISC-V Matrix Rotation — Assembly & Memory-System Analysis **February 2024**

- Implemented multiple in-place matrix rotation algorithms in RISC-V assembly, including transpose, layer-by-layer, and divide-and-conquer approaches
- Analyzed instruction and memory traces to evaluate cache behavior, locality, and architectural tradeoffs, using results to recommend an optimized cache configuration.

SKILLS & TOOLS

- **Programming:** C, C++, Python, Verilog, VHDL, RISC-V Assembly, MATLAB
- **Embedded Systems:** Embedded C, ESP32, STM32, PSoC, FreeRTOS, I2C, SPI, UART, GPIO, ADC, Timers, Interrupts
- **Hardware/PCB:** KiCad, Altium, Cadence Virtuoso, LTSpice, PCB Layout, Schematic Design, Soldering, Oscilloscope, Multimeter, Logic Analyzer
- **Digital Design:** Verilog RTL, FSM Design, Testbenches, ModelSim, Intel Quartus Prime, FPGA Synthesis